Cover census over cubic extensions: the minimal GHS genus is set by `#E(K) mod 4`

2026-10-12. Conductor T-2261. **Structural stage diagnostic on toy fields.** It counts covers by Riemann–Hurwitz genus. No curve over `F_q` is constructed, no transfer map is evaluated, no discrete logarithm is solved, and no speedup is claimed. Protocol written before the first run: PROTOCOL.md (protocol-versioned at commit, not independently preregistered; the 2026-10-13 amendment adds references and this status). Current run `20261013T0403Z`: records, environment (carries the sha256 of the source and of PROTOCOL.md), run report. It re-ran the same deterministic tool after the tool gained aggregate `mod4` records and a neighbor-model count; the per-curve records are identical to the earlier run `20261013T0305Z` (records, environment, run report), which is kept unchanged as superseded evidence. This report as PDF: REPORT.pdf. Follows `research/weak_subfield_hyperelliptic_decomposition_20261012/` (PR #1757), which showed that same-field higher-genus representations do not help.

Question

For `E` over `K = F_{q³}`, what is the smallest genus of a curve over `F_q` that the odd-characteristic GHS construction attaches to `E`, over every `F_q`-structure on the `x`-line? What changes after a 2-isogeny, and over the whole isogeny class?

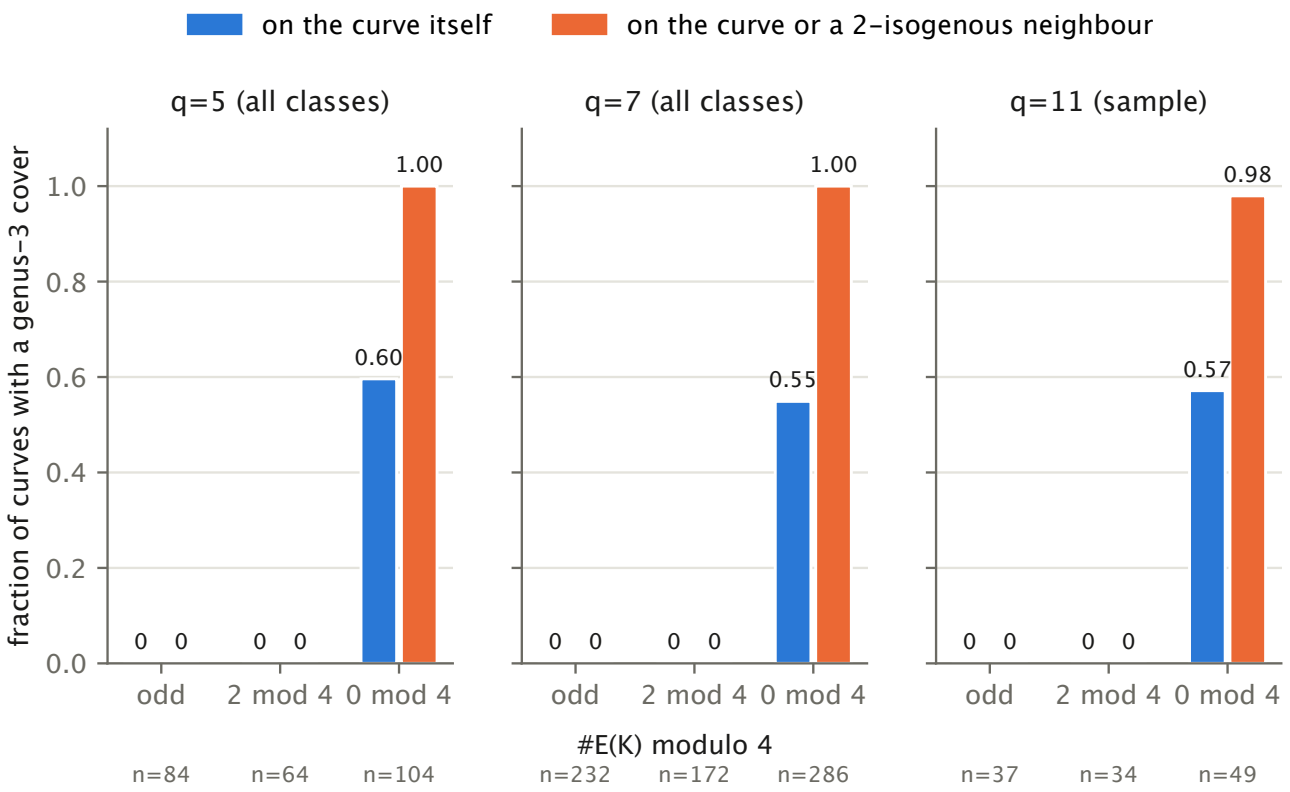
Result

With one exception the minimal genus over the isogeny class follows `#E(K) mod 4` (counts are isomorphism classes of curves):

`#E(K)`	minimal genus over the isogeny class	status
odd	13	derived (below) and measured, 316 of 316 curves at q = 5, 7 and 37 of 37 sampled at q = 11
2 mod 4	5 (one exception: 9)	measured: 5 on 234 of 236 curves at q = 5, 7, 9 on the other two (one class at q = 5); never 3
0 mod 4	3	measured, 390 of 390 curves at q = 5, 7

Genus 3 is the only value below the rho exponent, `q^{3/2}` (Pollard 1978), because index calculus on a genus-`g` Jacobian costs `Õ(q^{2−2/g})` (Diem, Math. Comp. 80 (2011), a theorem for each fixed `g ≥ 2`), which is `q^{4/3}` at `g = 3` and `q^{3/2}` at `g = 4`. The special `~q` figure for non-hyperelliptic genus 3 (plane quartics) is a heuristic of Diem–Thomé, J. Cryptology 21 (2008), not a theorem. Full references with DOIs: PROTOCOL.md, 2026-10-13 amendment.

So on the exhaustively censused fields `q = 5, 7` a genus-3 cover exists over the isogeny class exactly for the classes with `4 | #E(K)` (390 of 390 such curves have one; none of the 316 odd-order and 236 order-2-mod-4 curves does), and a curve of prime order, or of order twice an odd number, has none. At `q = 11` only a sample of 120 curves was run and class minima were not computed: all 37 odd and 34 “2 mod 4” sampled curves stay at genus at least 5, but of 49 sampled curves with `4 | #E(K)`, 48 reach genus 3 within one 2-isogeny and **one does not** (its minimum within one step is 5; whether a longer isogeny path reaches 3 was not computed). The “exactly” is therefore a statement about `q = 5, 7`; at `q = 11` it is a sample with one exception. None of the rows is a theorem except the odd row.



Measurements (run 20261013T0305Z, exact counts)

Tool: cover_census.rs. For each curve it runs through every cocycle `A` (the tool found exactly `(Q³−Q)/(q³−q)` of them in each field: 16275, 120099, 1786323). The q = 11 arm computed 120 primary models plus 127 neighbor models (the K-rational 2-isogenous neighbours of the sampled curves, `totals` record: `neighbor_models` = 127, `models_total` = 247), each over all 1786323 structures. The genus formula agreed with the sum of the genera of the quadratic subcovers on every structure where it was checked (genus at most 5, or at the running minimum): 0 mismatches.

Curves by minimal genus, split by `#E(K) mod 4`. "self" is the curve, "≤1 step" the minimum over the curve and its K-rational 2-isogeny neighbours, "class" the minimum over all curves with the same `#E(K)`.

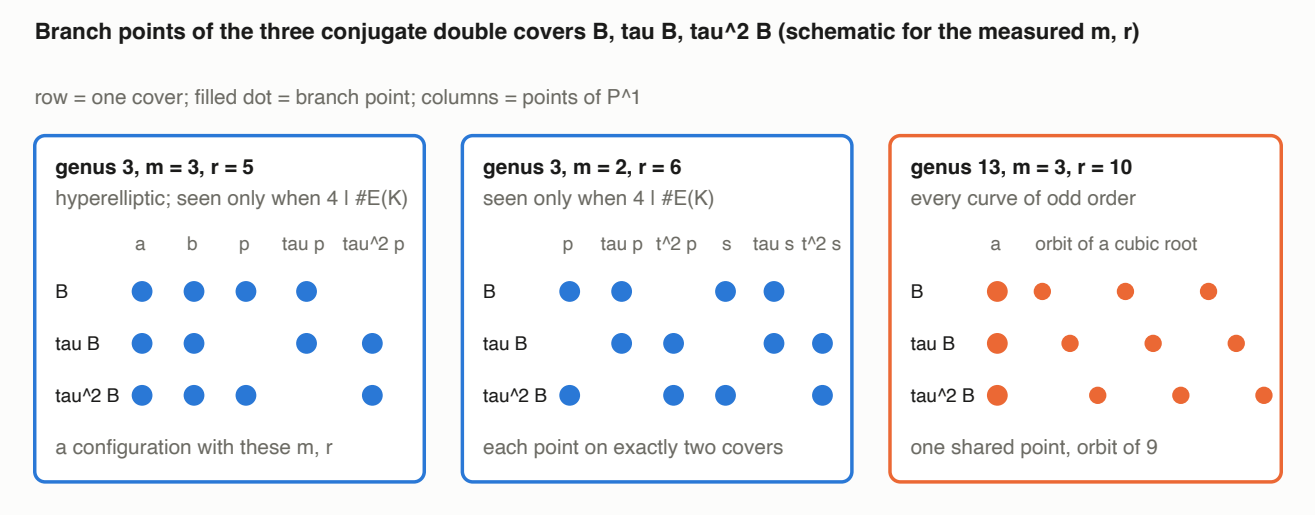
field	#E(K)	curves	self: genus 3 / 5 / 9 / 13	≤1 step: genus 3 / 5 / 9 / 13	class: genus 3 / 5 / 9 / 13
q = 5, all 252 classes	odd	84	0 / 0 / 0 / 84	0 / 0 / 0 / 84	0 / 0 / 0 / 84
	2 mod 4	64	0 / 60 / 4 / 0	0 / 60 / 4 / 0	0 / 62 / 2 / 0
	0 mod 4	104	62 / 42 / 0 / 0	104 / 0 / 0 / 0	104 / 0 / 0 / 0
q = 7, all 690 classes	odd	232	0 / 0 / 0 / 232	0 / 0 / 0 / 232	0 / 0 / 0 / 232
	2 mod 4	172	0 / 168 / 4 / 0	0 / 168 / 4 / 0	0 / 172 / 0 / 0
	0 mod 4	286	157 / 126 / 3 / 0	286 / 0 / 0 / 0	286 / 0 / 0 / 0
q = 11, sample of 120	odd	37	0 / 0 / 0 / 37	0 / 0 / 0 / 37	not computed
	2 mod 4	34	0 / 33 / 1 / 0	0 / 33 / 1 / 0	not computed
	0 mod 4	49	28 / 21 / 0 / 0	48 / 1 / 0 / 0	not computed

At `q = 5` two curves of order 2 mod 4 sit in an isogeny class whose minimum is 9; at `q = 7` every such class reaches 5.

Types at the minimum (number of curves; a curve can have both genus-3 types):

field	genus 3, `m=2, r=6` only	both	genus 3, `m=3, r=5` only (hyperelliptic)	genus 5, `m=3, r=6` (hyperelliptic)	genus 9, `m=3, r=8`	genus 13, `m=3, r=10`
q = 5	18	24	20	102	4	84
q = 7	66	49	42	294	7	232
q = 11 (sample)	18	4	6	54	1	37

A genus-3 curve has between 2 and 18 structures attaining it at `q = 5` and between 2 and 24 at `q = 7`, out of 16275 and 120099. The smallest cofactor among curves with a genus-3 cover is 4 (`#E(K) = 124 = 4·31` at `q = 5`, `332 = 4·83` at `q = 7`).



Derivation for odd order

Proposition (derived; orbit step in Lean). If `#E(K)` is odd, every structure gives genus at least 13, and 13 is attained.

Proof. The branch set is one `K`-rational point `a` and the three roots of a cubic irreducible over `K`. A root `θ` has `τ⁹θ = θ` and `τ³θ ≠ θ` (`τ³` is the Frobenius of `K`), so `τ^kθ ≠ θ` for `1 ≤ k ≤ 8` (`OrbitNine.no_short_return`) and the nine iterates are distinct. They are all branch points of some `τ^i B`, so `r ≥ 9 + 1`, with equality iff `τa = a`. No two of `B, τB, τ²B` coincide and their triple symmetric difference contains `θ`, so `m = 3` and `g = 2(r − 4) + 1 ≥ 13`. ∎

The same holds for every curve isogenous over `K`, since the order is an isogeny invariant. P2 of the protocol predicted "at least 5"; the proof and the data give 13.

Lean: lean/OrbitNine.lean, toolchain v4.34.1, no Mathlib; `#print axioms` reports `propext` only.

Report statement	Lean declaration	Status
no return of `θ` before 9 steps	`no_short_return`	proved
`τ³` is the Frobenius of `K`; branch set; genus formula	none	**not formalized**

No identity certificate: the report claims no polynomial identity.

What this does and does not say

- **Prime-order curves over a cubic extension have no small cover of this kind**, and no isogeny changes that. The route needs cofactor at least 4.
- **For `4 | #E(K)` a genus-3 cover is one 2-isogeny away** on 438 of 439 curves examined (104 at q = 5 and 286 at q = 7, exhaustive; 48 of the 49 sampled at q = 11); the remaining `q = 11` curve was not followed further.
- **Not established:** that the genus-3 curve has a model over `F_q` with the map `E(K) → Jac(C)(F_q)` injective on the odd part (standard for odd `n`, cited from memory, not verified here); whether the `m = 2` curves are plane quartics; any cost. The exponents in PROTOCOL.md are published asymptotics, not measurements.
- **Scope:** `n = 3`, `q ∈ {5, 7, 11}`, odd characteristic above 3, the `(2,…,2)` construction only. Other covers (trigonal, `n = 5, 7`, characteristic 2) are not covered. A bounded census is not a proof for all `q`.

Next discriminating step

Build the genus-3 curve over `F_q` for one `q = 7` class of order `4·ℓ` and check the transfer on the order-`ℓ` subgroup. If it is injective, a genus-3 index-calculus count against rho on that toy instance is the first measurement that could say anything about cost. If the `F_q`-model fails to exist, the census counts something the attack cannot use.

Graphs checked and left unchanged

No canonical graph changes (no ECDLP rung, no timing; no `benchmarks/` entry).

Reproduce

```
rustc --edition 2021 -O cover_census.rs && ./cover_census 5 && ./cover_census 7 && ./cover_census 11 120
python3 reports/plot.py results/run-20261013T0403Z.jsonl reports/run-20261013T0403Z   # renders recorded
`mod4` values only
cd lean && lake build
```